MSSC 6250 – Statistical Machine Learning
Midterm Examination — Solution Key

Total: 100 points

Problem 1 — Statistical Learning Framework (15 pts)

Suppose $Y = f(X) + \varepsilon$, with $\mathbb{E}(\varepsilon) = 0$.

(a) **(6 pts) Prediction vs inference; reducible vs irreducible error.**

- **Prediction:** Goal is accurate prediction of Y for new/unseen inputs X. We focus on minimizing expected test loss (e.g., MSE or misclassification error). The form/interpretability of $\hat{f}$ may be secondary (a "black box" can be acceptable).
- **Inference:** Goal is understanding the relationship between Y and predictors X (which predictors matter, direction/magnitude of effects, uncertainty via SEs/CIs/tests). Interpretability is important.
- **Reducible error:** Error due to estimating f with an approximation $\hat{f}$; can potentially be reduced by better modeling choices, more data, feature engineering, regularization, etc.
- **Irreducible error:** Error due to inherent randomness/noise ε (unmeasured factors). Cannot be eliminated even if f were known.

(b) **(5 pts) Show the decomposition.**
For fixed x,

$$\mathbb{E}\big[(Y - \hat{f}(x))^2\big] = \mathbb{E}\big[(f(x) + \varepsilon - \hat{f}(x))^2\big].$$

Expand:

$$= \mathbb{E}\Big((f(x) - \hat{f}(x))^2 + 2(f(x) - \hat{f}(x))\varepsilon + \varepsilon^2\Big).$$

Since $f(x) - \hat{f}(x)$ is constant given x and $\mathbb{E}(\varepsilon) = 0$,

$$= (f(x) - \hat{f}(x))^2 + 2(f(x) - \hat{f}(x))\mathbb{E}(\varepsilon) + \mathbb{E}(\varepsilon^2) = (f(x) - \hat{f}(x))^2 + \mathrm{Var}(\varepsilon).$$

Meaning: the first term is the *reducible* part (error from imperfect $\hat{f}$), and the second term is *irreducible* noise.

(c) **(4 pts) Bias–variance tradeoff.**
As model flexibility increases:

- **Bias decreases** (model can better approximate complex f).
- **Variance increases** (model becomes more sensitive to the specific training sample).

Consequently, test error often decreases initially (bias reduction dominates) and then increases (variance dominates / overfitting), producing a U-shaped relationship versus flexibility.

Problem 2 — Linear Regression Interpretation (25 pts)

Two models:

$$\text{Model 1: } Y = \beta_0 + \beta_1\,\text{AIUse} + \varepsilon, \qquad \text{Model 2: } Y = \beta_0 + \beta_1\,\text{AIUse} + \beta_2\,\text{GPA} + \varepsilon.$$

(a) **Interpret AIUse in Model 1.**
AIUse is binary. The estimate $\hat{\beta}_1 = +6$ means:

$$\widehat{\mathbb{E}}[Y \mid \text{AIUse} = 1] - \widehat{\mathbb{E}}[Y \mid \text{AIUse} = 0] \approx 6.$$

So students who regularly use AI tools are predicted to score **about 6 points higher** on the final exam *on average* than students who do not, *without adjusting for other variables.* The association is statistically significant (p=0.003).

(b) **Interpret AIUse in Model 2 and compare to Model 1.**
In Model 2, $\hat{\beta}_1 = -3$ means:

$$\widehat{\mathbb{E}}[Y \mid \text{AIUse} = 1, \text{GPA} = g] - \widehat{\mathbb{E}}[Y \mid \text{AIUse} = 0, \text{GPA} = g] \approx -3,$$

i.e., **holding GPA fixed**, regular AIUse is associated with **about 3 points lower** exam score on average. This differs from Model 1 because Model 1 is *unadjusted* (marginal association), while Model 2 is *adjusted/conditional* on GPA.

(c) **Explain why the coefficient changes from $+6$ to -3 after including GPA.**
This is consistent with **omitted variable bias** in Model 1. GPA is a strong predictor of exam score (Model 2 shows $\hat{\beta}_2 = +12$, highly significant). The sign flip suggests AIUse is associated with GPA (likely negatively: students with lower GPA may be more likely to rely on AI tools). When GPA is omitted, the AIUse coefficient partially captures differences in GPA between AIUse groups, yielding a misleading positive association. After adjusting for GPA, AIUse reflects the association *net of GPA*, which can differ in magnitude and even change sign.

(d) **Which model provides a more reliable estimate and why?**
Model 2 is more reliable for the association between AIUse and exam score because it adjusts for prior academic preparation (GPA), which strongly predicts performance and plausibly differs between AIUse groups. (Still observational: causality is not guaranteed; other confounders may remain.)

Problem 3 — Formulation of Classification Methods (20 pts)

Suppose $Y \in \{0, 1\}$ and $X \in \mathbb{R}^p$.

(a) **Logistic Regression.**
Model:

$$P(Y = 1 \mid X = x) = \frac{e^{\beta_0 + \beta^T x}}{1 + e^{\beta_0 + \beta^T x}}.$$

Log-odds (logit) form:

$$\log\left(\frac{P(Y = 1 \mid x)}{1 - P(Y = 1 \mid x)}\right) = \beta_0 + \beta^T x.$$

Parameters (β_0, β) are estimated by **maximum likelihood** (maximize Bernoulli log-likelihood), typically via Newton–Raphson / IRLS.

(b) **LDA.**
Distributional assumption:

$$X \mid Y = k \sim \mathcal{N}(\mu_k, \Sigma), \quad k \in \{0, 1\}.$$

Key covariance assumption: **common covariance matrix** Σ for both classes $\Rightarrow$ linear decision boundary. Estimated quantities: class priors $\pi_k = P(Y = k)$, class means μ_k, and pooled covariance matrix Σ.

(c) **QDA.**
Distributional assumption:

$$X \mid Y = k \sim \mathcal{N}(\mu_k, \Sigma_k), \quad k \in \{0, 1\}.$$

Difference from LDA: each class has its own covariance matrix Σ_k. Effect: decision boundary becomes **quadratic** (more flexible than LDA; typically higher variance due to more parameters).

(d) **Naive Bayes.**
Factorization:

$$p(X \mid Y = k) = \prod_{j=1}^{p} p(X_j \mid Y = k).$$

Assumption: **conditional independence** of features given the class Y. This reduces parameters because we estimate only **univariate** class-conditional distributions for each feature rather than a full joint distribution/covariance structure.

Problem 4 — Resampling & Cross-Validation (20 pts)

(a) **K-fold CV procedure.**
Split data into K roughly equal folds. For $k = 1, \dots, K$: fit the model on the other $K - 1$ folds, predict on fold k, compute the validation error on fold k (e.g., MSE or misclassification rate). Average the K validation errors to get the CV estimate of test error.

(b) **LOOCV vs 10-fold (bias/variance).**
LOOCV uses $K = n$ folds:

- **Lower bias** (training set is size $n - 1$, close to full data).
- Often **higher variance** (validation errors are highly correlated; each training set is almost identical).

10-fold CV:

- Slightly **higher bias** than LOOCV (smaller training sets).
- Typically **lower variance** and better practical tradeoff; also less computation.

(c) **Why training error underestimates test error.**
Training error is evaluated on the same data used to fit the model, so it benefits from overfitting and adapting to noise in the training set. Therefore it is optimistically biased and typically smaller than the true generalization (test) error.

(d) **Why CV useful for tuning parameters.**
Tuning parameters (e.g., polynomial degree, λ in ridge/lasso, K in KNN) control model complexity and the bias–variance tradeoff. CV provides an estimate of test error for each candidate value; we choose the tuning parameter that minimizes estimated test error, improving generalization and reducing overfitting.

Problem 5 — Model Selection & Regularization (20 pts)

(a) **Best subset vs forward stepwise.**

- **Best subset:** considers all 2^p subsets of predictors (or all $\binom{p}{k}$ subsets of size k for each k). Computationally expensive due to exponential growth in the number of models.
- **Forward stepwise:** greedy procedure; starts with null model and adds one predictor at a time based on improvement criterion, fitting far fewer models (roughly $O(p^2)$ fits).

(b) **Why least squares variance increases when $n \approx p$ or $n < p$.**
OLS $\hat{\beta} = (X^T X)^{-1} X^T y$ (when invertible).

- If $n \approx p$, $X^T X$ can be ill-conditioned; $(X^T X)^{-1}$ can have large entries, leading to large variance in $\hat{\beta}$ and unstable predictions.
- If $n < p$, $X^T X$ is singular (not invertible) and OLS is not uniquely defined; coefficients are not identifiable and solutions can be highly unstable without regularization.

(c) **Ridge regression and role of λ.**
Ridge solves:

$$\min_{\beta} \left\{ \sum_{i=1}^{n} (y_i - x_i^T \beta)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 \right\}.$$

$\lambda \geq 0$ controls the strength of shrinkage:

- $\lambda \to 0$: ridge approaches OLS.
- $\lambda \to \infty$: coefficients shrink toward 0 (intercept typically unpenalized).

Ridge increases bias but reduces variance, often improving test performance when OLS is high-variance.

(d) **Ridge vs lasso; variable selection.**

- **Lasso** (ℓ_1 penalty) can set some coefficients exactly to zero $\Rightarrow$ performs variable selection.
- **Ridge** (ℓ_2 penalty) shrinks coefficients but typically does not set them exactly to zero.

Reason: the geometry of the ℓ_1 penalty encourages sparse solutions (corners), while ℓ_2 is smooth.